



AQ1.3: Activity Questions 3 - Not Graded



This assignment will not be graded and is only for practice.

3.1 Level 1:

Consider a system of linear equations (System 1):

$$\begin{aligned} -2x_1 + 3x_2 + x_3 &= 1 \\ -x_1 + x_3 &= 0 \\ 2x_2 &= 5 \end{aligned}$$

Answer questions 1 and 2 based on the above data.

1 point

If the matrix representation of system (1) is $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then



$$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$$



$$b = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$$



$$A = \begin{bmatrix} -2 & -10 \\ 3 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$$



$$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$$

Yes, the answer is correct.

Score: 1

Targeted Feedback:

Feedback:

Accepted Answers:



$$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix} A = \begin{bmatrix} -2 & -10 & 3 & 0 & 2 & 1 & 1 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix} A = \begin{bmatrix} -2 & -10 & 3 & 0 & 2 & 1 & 1 & 0 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix} b = \begin{bmatrix} 1 & 0 & 5 \end{bmatrix}$$

1 point

System (1) has

[Hint: Solve for x_1, x_2, x_1, x_2 , and x_3, x_3 .]

- ☒ a unique solution.
☐ no solution.
☐ infinitely many solutions.
☐ None of the above.

Yes, the answer is correct.

Score: 1

Accepted Answers:

a unique solution.

1 point

Choose the set of correct options.

[Hint: Think of AA as a 2×2 or 3×3 matrix, and bb accordingly.]

- ☒ Every system of linear equations has either a unique solution, no solution or infinitely many solutions.
☐

If each equation of a system of linear equations is multiplied by a non-zero constant c , then the solution of the new system of equations is c times the solution of the old system of equations.



If $Ax = b$ is a system of linear equations which has a solution, then the system of linear equations $cAx = bcAx = b$, where $c \neq 0$, will also have a solution.



If $Ax = b$ is a system of linear equations which has a solution, then $\frac{1}{c}Ax = \frac{1}{c}b$, where $c \neq 0$, will also have a solution.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Every system of linear equations has either a unique solution, no solution or infinitely many solutions.

If $Ax = b$ is a system of linear equations which has a solution, then the system of linear equations $cAx = bcAx = b$, where $c \neq 0$, will also have a solution.

If $Ax = b$ is a system of linear equations which has a solution, then $\frac{1}{c}Ax = \frac{1}{c}b$, where $c \neq 0$, will also have a solution.

1 point

The Plane 1 and Plane 2 in Figure M2W1AQ3, correspond to two different linear equations, which form a system of linear equations.

The above system of linear equations has

[Hint: If a system of linear equations has a solution, then the point corresponding to the solution must lie on each plane corresponding to each linear equation of the given system.]

- ☐ a unique solution.
☒ no solution.
☐ infinitely many solutions.
☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

no solution.

1 point

Consider the geometric representations (Figures (a), (b), (a), (b), and (c(c))) of three systems of linear equations.

Choose the set of correct options.



Figure (a)(a) represents a system of linear equations which has no solution.



Figure (a)(a) represents a system of linear equations which has infinitely many solutions.



Figure (b)(b) represents a system of linear equations which has a unique solution.



Figure (b)(b) represents a system of linear equations which has infinitely many solutions.



Figure (c)(c) represents a system of linear equations which has infinitely many solutions.



Figure (c)(c) represents a system of linear equations which has no solution.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Figure (a)(a) represents a system of linear equations which has no solution.

Figure (b)(b) represents a system of linear equations which has a unique solution.

Figure (c)(c) represents a system of linear equations which has infinitely many solutions.

3.2 Level 2:

1 point

Consider a system of equations:
$$\begin{aligned} 2x_1 + 3x_2 &= 6 \\ -2x_1 + kx_2 &= d \\ 4x_1 + 6x_2 &= 12 \end{aligned}$$
 Choose

the set of correct options.



$Ax = b$ represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$

and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$



The system has no solution if $k = -3$, $d = 0$.



The system has a unique solution if $k = 3$, $d = 0$.



The system has infinitely many solutions if $k = -3$, $d = 6$.



The system has infinitely many solutions if $k = -3$, $d = -6$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$Ax = b$ represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$, $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$,

and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$

The system has no solution if $k = -3$, $d = 0$.

The system has a unique solution if $k = 3$, $d = 0$.

The system has infinitely many solutions if $k = -3$, $d = -6$.

1 point

Let x_1 and x_2 be solutions of the system of linear equations $Ax = b$. Which of the following options are correct?



$x_1 + x_2$ is a solution of the system of linear equations $Ax = b$.



$x_1 + x_2$ is a solution of the system of linear equations $Ax = 2b$.



$x_1 - x_2$ is a solution of the system of linear equations $Ax = b$.



$x_1 - x_2$ is a solution of the system of linear equations $Ax = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x_1 + x_2$ is a solution of the system of linear equations $Ax = 2b$.

$x_1 - x_2$ is a solution of the system of linear equations $Ax = 0$.

1 point

Let v be a solution of the systems of linear equations $A_1x = b$ and $A_2x = b$. Which of the following options are correct?



v is a solution of the system of linear equations $(A_1 + A_2)x = b$.



v is a solution of the system of linear equations $(A_1 + A_2)x = 2b$.



v is a solution of the system of linear equations $(A_1 - A_2)x = 0$.



v is a solution of the system of linear equations $(A_1 - A_2)x = b$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

v is a solution of the system of linear equations $(A_1 + A_2)x = 2b$.

v is a solution of the system of linear equations $(A_1 - A_2)x = 0$.

Consider a system of equations: $\begin{cases} x_1 - 3x_2 = 4 \\ 3x_1 + kx_2 = -12 \end{cases}$ Where $k \in \mathbb{R}$.

If the given system has a unique solution, then k should not be equal to

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -9

1 point

Consider two system of linear equations:

System 1:

$$\begin{cases} x_1 + 2x_2 = -5 \\ 0x_1 - x_2 = 5 \end{cases} \Rightarrow x_1 + 2x_2 = -5 \Rightarrow x_1 - x_2 = -5 \Rightarrow 5$$

System 2:

$$\begin{cases} -2x_3 + x_4 = -5 \\ 3x_3 + x_4 = 5 \end{cases} \Rightarrow -2x_3 + x_4 = -5 \Rightarrow 3x_3 + x_4 = -5 \Rightarrow 5$$

Suppose there is another system of linear equation given by

$$\begin{cases} (1-2)(x_1 + x_3) + (2+1)(x_2 + x_4) = m \\ (0+3)(x_1 + x_3) + (-1+1)(x_2 + x_4) = n \end{cases} \Rightarrow \begin{cases} (1-2)(x_1 + x_3) + (2+1)(x_2 + x_4) = m \\ (0+3)(x_1 + x_3) + (-1+1)(x_2 + x_4) = n \end{cases}$$

$$(-1+1)(x_2 + x_4) = m = n$$

for some real values of m and n . Find the value of $n - m$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 46

1 point